

PORTFOLIO PROJECT REPORT

E-COMMERCE SQL BUSINESS ANALYTICS QUERY LIBRARY

SQL Analytics & Excel Dashboard | Data Analyst Portfolio

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Executive Summary

This report documents a SQL business-analytics query library built against an e-commerce database (Customers, Orders, Order_Items, Products). The project contains 52 standalone queries answering real business questions spanning revenue, product performance, customer segmentation, geography, order behavior, and growth trends.

The query set progresses from simple aggregations (total revenue, order counts) to advanced analytical patterns — window functions, running totals, cohort-style acquisition tracking, RFM segmentation, churn detection, and market-basket (product affinity) analysis — reflecting the kind of analytical depth expected of a working Data Analyst.

AI tools were used to accelerate query drafting; every query was reviewed, understood, and validated by the analyst against the schema and underlying data before being finalized. Section 2 details this working approach. Section 8 documents a companion deliverable built on the same dataset: a single-page Excel dashboard (PivotTables, PivotCharts, and slicers) designed for ongoing, self-serve monitoring rather than one-off queries.

1. Project Overview

This project is a structured library of MySQL queries designed to answer the kinds of operational and strategic questions a real e-commerce business would ask of its data — from "What is our total revenue?" to "Which customers are at risk of churn?" and "What is each customer's RFM segment?"

Objective

- Translate common e-commerce business questions into accurate, efficient SQL queries.
- Demonstrate the ability to work across the full SQL toolkit — joins, aggregations, subqueries, CTEs, and window functions.
- Build reusable analytical patterns (RFM, churn, running totals, growth rates) applicable to real business reporting.
- Validate every query's logic and output against the source data rather than trusting generated SQL blindly.

Tools & Skills Used

- **Core platform:** MySQL
- **Query Techniques:** Multi-table JOINS (Customers, Orders, Order_Items, Products)
- Aggregate functions — SUM, COUNT, AVG, ROUND
- Window functions — ROW_NUMBER, DENSE_RANK, NTILE, LAG, LEAD, running SUM() OVER()
- Common Table Expressions (CTEs), including multi-step and nested CTEs
- Conditional aggregation with CASE WHEN for segmentation and rate calculations
- Self-joins for market-basket / product-affinity analysis
- Date functions — DATE_FORMAT, DATEDIFF, DATE_SUB for trend and recency analysis
- **Business Analysis:** Business framing of technical queries: revenue, churn, CLV, RFM, growth-rate analysis

2. Development Approach: AI-Assisted, Analyst-Validated

In the interest of transparency, this project's development workflow is documented below as part of standard professional practice:

"I used AI to accelerate query development, but I understand the schema, joins, filters, aggregations and business logic, and I validate the generated query against the data."

In practice, this meant using AI as a drafting accelerator — particularly for boilerplate JOIN structures and window-function syntax — while the analyst retained full ownership of: translating each business question into the correct logical approach, selecting the appropriate joins and filters (e.g., consistently scoping revenue metrics to `order_status = 'Delivered'`), verifying aggregation grain (per-order vs. per-line-item vs. per-customer), and manually checking query outputs against the underlying tables for correctness.

This mirrors how SQL is increasingly used in real analytics roles: AI accelerates syntax and first drafts, while the analyst is accountable for correctness, business logic, and interpretation — the skills this project is intended to demonstrate.

3. Database Schema

The query library runs against four related tables. Below are the columns referenced throughout the queries, grouped by table:

Customers

Column	Used For
customer_id	Primary key; joined to Orders
customer_name	Customer identification in reporting
city	Geographic revenue analysis
gender	Revenue segmentation by gender

Orders

Column	Used For
order_id	Primary key; joined to Order_Items
customer_id	Foreign key to Customers
order_date	Time-series, recency, and trend analysis
order_status	Filtering (e.g., 'Delivered', 'Cancelled')

Order_Items

Column	Used For
order_id	Foreign key to Orders
product_id	Foreign key to Products
quantity	Revenue and volume calculations
unit_price	Revenue calculations (quantity × unit_price)

Products

Column	Used For
product_id	Primary key; joined to Order_Items
product_name	Product-level reporting
category	Category-level aggregation
unit_price	Pricing analysis (e.g., most expensive product sold)

4. Business Questions Addressed (52 Queries)

The 52 queries are organized below into seven functional categories, mirroring how a business stakeholder would group these questions.

4.1 Revenue & Sales Overview (8 queries)

#	Business Question
1	What is our total revenue so far?
2	How many total orders have we received?
3	How many total customers do we have?
4	How many total products do we sell?
5	What is the average amount a customer spends per order?
9	How has revenue trended month by month?
41	What is the daily sales trend?
50	Which month had the highest revenue?

4.2 Product Performance (12 queries)

#	Business Question
6	What are the top 5 best-selling products?
7	Which category is generating the most revenue?
14	What is the best-selling category?
15	Which product generates the highest revenue?
16	Which product generates the lowest revenue?
22	What is the top-selling product in each category?
28	What percentage does each product contribute to total revenue?
36	Which two products are most frequently purchased together?
42	What are the top 10 best-selling products?
48	What is the worst-selling category?
49	What is the most expensive product that was sold?
52	What percentage does each category contribute to total revenue?

4.3 Customer Analytics & Segmentation (17 queries)

#	Business Question
8	Who are our top 5 most valuable customers?
11	Which customers purchase repeatedly?
12	Which customers have never placed an order?
13	Which customers have been inactive in the last 90 days?
19	Which customers have the highest average order value?
21	How are customers ranked based on revenue?
27	Who are the top 3 customers by revenue?
30	What was each customer's highest-value order?

31	What is each customer's lifetime value (CLV)?
32	What percentage of customers make repeat purchases?
33	How many customers are new vs returning?
35	Which customers purchase from more than one category?
39	Which customers are at risk of churn?
40	RFM Analysis — Recency, Frequency, Monetary segmentation
43	Who are the top 10 customers by revenue?
44	How much revenue comes from male vs female customers?
45	What is our customer churn rate?

4.4 Geographic Analysis (2 queries)

#	Business Question
10	Which city generates the most sales?
20	What are the top 3 cities by revenue?

4.5 Order Behavior & Timing (6 queries)

#	Business Question
17	On average, how many products are bought per order?
18	Which orders are worth more than 10,000?
24	When was each customer's previous order placed?
25	When was each customer's next order placed?
26	How many days are there between a customer's consecutive orders?
37	On average, how many days pass between a customer's orders?

4.6 Growth & Trend Analysis (5 queries)

#	Business Question
23	How has revenue accumulated over time (running total)?
29	What is the month-over-month revenue growth?
34	How many customers were active each month?
38	Which product category is growing the fastest?
51	How many new customers were acquired each month?

4.7 Order Status Analysis (2 queries)

#	Business Question
46	What percentage of orders get cancelled?
47	How many orders were delivered vs cancelled?

5. Highlighted Advanced Queries

A sample of the more advanced patterns from the library, chosen to demonstrate range beyond basic aggregation:

5.1 Top-Selling Product per Category (Window Function + CTE)

```
WITH Product_Sales AS (
    SELECT p.category, p.product_name, SUM(oi.quantity) AS Total_Quantity
    FROM products p
    JOIN order_items oi ON p.product_id = oi.product_id
    JOIN orders o ON oi.order_id = o.order_id
    WHERE o.order_status = 'Delivered'
    GROUP BY p.category, p.product_id, p.product_name
),
Ranked_Products AS (
    SELECT *, ROW_NUMBER() OVER (
        PARTITION BY category ORDER BY Total_Quantity DESC
    ) AS rn
    FROM Product_Sales
)
SELECT category, product_name, Total_Quantity
FROM Ranked_Products WHERE rn = 1;
```

Uses ROW_NUMBER() with PARTITION BY to find the #1 product within each category in a single pass, avoiding a correlated subquery per category.

5.2 Running Total of Revenue Over Time

```
WITH Daily_Revenue AS (
    SELECT o.order_date, SUM(oi.quantity * oi.unit_price) AS Daily_Revenue
    FROM orders o
    JOIN order_items oi ON o.order_id = oi.order_id
    WHERE o.order_status = 'Delivered'
    GROUP BY o.order_date
)
SELECT order_date, ROUND(Daily_Revenue, 2) AS Daily_Revenue,
    ROUND(SUM(Daily_Revenue) OVER (ORDER BY order_date), 2)
    AS Running_Total_Revenue
FROM Daily_Revenue ORDER BY order_date;
```

A cumulative SUM() OVER (ORDER BY ...) window function produces a running revenue total without a self-join or procedural loop.

5.3 RFM Segmentation (Recency, Frequency, Monetary)

```
WITH Customer_RFM AS (
    SELECT c.customer_id, c.customer_name,
        DATEDIFF((SELECT MAX(order_date) FROM orders),
            MAX(o.order_date)) AS Recency,
        COUNT(DISTINCT o.order_id) AS Frequency,
        SUM(oi.quantity * oi.unit_price) AS Monetary
    FROM customers c
    JOIN orders o ON c.customer_id = o.customer_id
    JOIN order_items oi ON o.order_id = oi.order_id
    WHERE o.order_status = 'Delivered'
    GROUP BY c.customer_id, c.customer_name
)
```

```
SELECT customer_id, customer_name, Recency, Frequency,
       ROUND(Monetary, 2) AS Monetary,
       NTILE(5) OVER (ORDER BY Recency DESC) AS Recency_Score,
       NTILE(5) OVER (ORDER BY Frequency) AS Frequency_Score,
       NTILE(5) OVER (ORDER BY Monetary) AS Monetary_Score
FROM Customer_RFM ORDER BY Monetary DESC;
```

Combines a CTE with three independent NTILE(5) window functions to score every customer 1–5 on Recency, Frequency, and Monetary value — the standard RFM customer-segmentation model used in retention marketing.

5.4 Frequently Purchased Together (Market Basket / Self-Join)

```
SELECT p1.product_name AS Product_1, p2.product_name AS Product_2,
       COUNT(DISTINCT oi1.order_id) AS Times_Purchased_Together
FROM order_items oi1
JOIN order_items oi2
     ON oi1.order_id = oi2.order_id
     AND oi1.product_id < oi2.product_id
JOIN products p1 ON oi1.product_id = p1.product_id
JOIN products p2 ON oi2.product_id = p2.product_id
JOIN orders o ON oi1.order_id = o.order_id
WHERE o.order_status = 'Delivered'
GROUP BY p1.product_id, p1.product_name, p2.product_id, p2.product_name
ORDER BY Times_Purchased_Together DESC LIMIT 1;
```

A self-join on order_items (with product_id < product_id to avoid double-counting pairs) identifies the most common product combination — a lightweight market-basket analysis useful for cross-sell recommendations.

6. SQL Techniques & Skills Demonstrated

- Multi-table INNER JOINS and LEFT JOINS (e.g., identifying customers with zero orders via LEFT JOIN ... WHERE ... IS NULL).
- CTEs, including chained and nested CTEs for multi-stage logic (e.g., month-over-month growth, RFM).
- Window functions: ROW_NUMBER, DENSE_RANK, NTILE, LAG, LEAD, and running SUM() OVER().
- Conditional aggregation with CASE WHEN for churn rate, cancellation rate, and new-vs-returning segmentation.
- Safe division using NULLIF() to prevent divide-by-zero errors in growth-rate calculations.
- Date arithmetic with DATE_FORMAT, DATEDIFF, and DATE_SUB for cohort, recency, and trend analysis.
- Consistent business-rule filtering (order_status = 'Delivered') applied across revenue-bearing queries to avoid overstating figures with cancelled orders.

7. Challenges & Learnings

- Ensured revenue-related queries consistently filtered to order_status = 'Delivered', since early draft queries risked double-counting cancelled orders as revenue.
- Validated window-function partitioning carefully (e.g., PARTITION BY customer_id for LAG/LEAD) to avoid mixing one customer's order history with another's.
- Used NULLIF() in growth-rate formulas after encountering divide-by-zero errors on months with no prior-period revenue.
- Cross-checked AI-generated query output against manual spot-checks of the raw tables to confirm joins were not silently duplicating rows (a common risk with multi-table joins on transactional data).

8. Excel Sales Dashboard

Alongside the SQL query library, the same e-commerce dataset was used to build a single-page interactive sales dashboard in Excel. Where the SQL library answers specific business questions on demand, the dashboard is built for ongoing monitoring: KPI cards, trend charts, and slicers that let a stakeholder filter by year, region, and customer segment without writing a query.

Dashboard Snapshot



Fig. 1 — E-Commerce Sales Analysis Dashboard, built in Excel (PivotTables, PivotCharts, and slicers over the Sample Superstore dataset)

Key Performance Indicators

KPI	Value & Trend
Sales	\$22,97,200.86 ▲ 20.62% vs. prior period
Profit	\$2,86,397.02 ▲ 14.41% vs. prior period
Quantity	37,873 units ▲ 28.64% vs. prior period
No. of Orders	9,994 orders ▼ 5.15% vs. prior period
Profit Margin	12.47% ▲ 27.45% vs. prior period

Each KPI card carries a small sparkline showing its month-by-month trajectory, so direction of travel is visible at a glance even before reading the headline number. The falling order count alongside rising sales and profit is flagged as a data point worth a closer look rather than a contradiction — it points to a rising average order value.

Visual Components

Panel	What It Shows
Sales and Profit Analysis	Combo chart (clustered columns for Sum of Sales, line for Sum of Profit) across Jan–Dec, showing seasonal peaks in Sep–Dec.
Category Wise Profit	Step/waterfall-style column chart showing cumulative profit build-up across categories (\$18.45K → \$122.49K → \$145.45K → \$286.40K).

Category Wise Sales Share %	Donut chart splitting revenue share across Technology (36.40%), Office Supplies (31.30%), and Furniture (32.30%).
Sales by State	Filled U.S. map with a graduated blue scale (\$0.92K to \$2,297.20K), highlighting the highest- and lowest-performing states at a glance.
Top 5 Subcategories by Sales	Horizontal bar chart ranking Phones, Chairs, Storage, Tables, and Binders — the five largest subcategories by revenue.
Filters	Slicers for Year (2011–2014), Region (Central, East, South, West), and Segment (Consumer, Corporate, Home Office) driving every visual on the page.

Excel Techniques & Skills Used

- PivotTables and PivotCharts as the data engine behind every card and chart on the dashboard.
- Slicers connected across multiple PivotTables via Report Connections, so one click filters the whole page.
- KPI cards built with linked cells and sparklines to show trend direction alongside the headline figure.
- Combo charts (column + line) for the Sales and Profit trend, and a donut chart for category share.
- Filled Map chart type for state-level sales, using Excel's built-in Bing/GeoNames geocoding.
- Custom dashboard theme: dark card backgrounds, consistent accent color, gridlines and cell borders removed for a clean, presentation-ready single page.
- Conditional formatting and custom number formats for currency, percentage, and up/down trend indicators.

Key Insights

- Technology is the top revenue-share category at 36.40%, narrowly ahead of Office Supplies (31.30%) and Furniture (32.30%) — the three categories are closely balanced rather than one dominating.
- Sales and profit both accelerate sharply from September through December, indicating a strong seasonal (holiday-quarter) pattern worth planning inventory and staffing around.
- Order count fell 5.15% even as sales, profit, and quantity all rose double digits — consistent with fewer, larger orders rather than a demand slowdown.
- Phones and Chairs lead all subcategories by sales, reinforcing Technology and Furniture as the categories to prioritize for promotions and stock planning.

9. Conclusion

Together, the SQL query library and the Excel dashboard demonstrate two complementary sides of the same e-commerce dataset: precise, validated SQL for ad-hoc and advanced analytical questions, and a PivotTable-driven Excel dashboard for ongoing, self-serve monitoring by non-technical stakeholders. Combined with an AI-assisted but analyst-owned development workflow, this reflects both technical proficiency and the judgment required to apply it correctly to real business questions — core competencies for an entry-level Data Analyst role.